What is Angular?

Angular is a **front-end web framework** developed by **Google**. It is used to build **dynamic, single-page web applications (SPAs)**.

- Written in **TypeScript** (a superset of JavaScript)
- Uses **components** to build UI
- Follows **MVC-like architecture**
- Comes with many built-in features (routing, forms, HTTP, etc.)

.html single web page
.html multiple pages website
Erp,pos , ms , webapplication , reactnative ,android app   vite (npm create vite@latest)

K

ey Features

1. **Component-Based Architecture**
   - The UI is built using components.
   - Each component has:
     - HTML template
     - CSS styles
     - TypeScript logic
2. **Two-Way Data Binding**
   - Automatically syncs data between UI and code.
3. **Dependency Injection**
   - Helps manage services and dependencies easily.
4. **Routing**
   - Navigate between different views/pages without reloading.
5. **Directives**
   - Extend HTML behavior.
   - Example:
     - `*ngIf` → show/hide elements
     - `*ngFor` → loop through lists
6. **Services**
   - Used to share data or logic across components.

Angular vs AngularJS

AngularJS (Old)	Angular (Modern)
Uses JavaScript	Uses TypeScript

AngularJS (Old)	Angular (Modern)
MVC architecture	Component-based
Slower	Faster & more optimized
Hard to maintain	Easy to scale

Basic Example

App Component

ng create c component-name

HTML (template):

```
<h1>{{ title }}</h1>
<button (click)="changeTitle()">Click Me</button>
```

TypeScript (logic):

```
import { Component } from '@angular/core';

@Component({
  selector: 'app-root',
  templateUrl: './app.component.html'
})
export class AppComponent {
  title = 'Hello Angular';

  changeTitle() {
    this.title = 'Angular is Awesome!';
  }
}
```

How Angular Works (Simple Flow)

1. User interacts with UI
2. Component handles logic
3. Data changes
4. UI updates automatically

Advantages

- Fast performance
- Strong typing (TypeScript)
- Large community
- Enterprise-level framework
- Built-in tools (CLI, routing, testing)

When to Use Angular

Use Angular if you want:

- Large-scale applications
- Structured code
- Strong architecture
- Enterprise projects

Tools Needed

- **Node.js**
- **Angular CLI** (Command Line Interface)

Install Angular CLI:

```
npm install -g @angular/cli
```

ng v

Create project:

```
ng new my-app
cd my-app
ng serve
```

Open in browser:

```
http://localhost:4200
```

Summary

Angular is a **powerful, full-featured framework** for building modern web applications.
It is best suited for **large and scalable projects**.